Bounded Recovery Structures of Linear Codes:
Transfer, Reliability, and Geometry

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Abstract

Fix a represented inner code I, a target x, and a helper bound r. Its normalized dual words through x of weight at most $r+1$ give the bounded recovery equations and their exact helper supports. For an outer family whose dual distance tends to infinity, we prove that all sufficiently long concatenations confine every such equation to its inner block if and only if $r+1 < z_x(I)$. Under this condition the supports and normalized coefficients are copied. Asymptotically good outer families therefore realize represented recovery data on a positive-density coordinate class over a fixed alphabet.

Two represented $[10,4,6]$ inner codes have the same pointed-perspective polynomial and matched locality, availability, transversal, blocker, and helper-degree statistics, but their radius-three reliabilities are

$$5s^3 - 7s^5 - s^6 + 5s^7 - s^9 \quad \text{and} \quad 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8.$$

A common outer family preserves matched global parameter bounds and places these recovery structures on target classes of density $1/10$. Thus these pointed-Tutte and conventional local invariants do not determine bounded stochastic repair behavior after positive-density realization.

We prove deletion–contraction and pivotal identities, a bounded BEC EXIT hierarchy whose differences recover the cheapest available repair radius, and reconstruction of MDS codes from minimum-weight recovery equations. Characteristic-three cubic–axis and quartic–nucleus examples exhibit geometric recovery structures.

1 Introduction

Locality asks how many surviving symbols suffice to recover one unavailable coordinate. Availability asks for many disjoint choices of helpers. Recovery sets and coordinate-indexed recovery structures are standard in locally repairable codes [7, 15, 13, 10], but locality and availability do not describe the whole local failure event: two codes can have the same values while their permitted recovery sets overlap very differently.

We use a pointed, coefficient-aware refinement of that standard data. For a target x, let $\mathcal{P}_x^{(\le r)}(C)$ be the dual words normalized by $w_x = 1$ and supported on x and at most r helpers. These are the normalized bounded recovery equations. Their exact helper supports generate the usual bounded recovery-set family by upward closure. The minimal supports measure disjoint availability and worst-case helper failure tolerance, while their monotone Boolean event gives exact stochastic reliability. The coefficients retain more: concatenation is controlled by the minimum cost of realizing the corresponding block functional rather than by support alone.

Theorem 1.1 (Main theorem: exact transfer and separated realization)**.** *Let $I \le \mathbb{F}_q^E$ be a fixed represented $[m,k,d]_q$ inner code, identify its message space with $L = \mathbb{F}_{q^k}$, let $x \in E$ be a repairable target, and put*

$$z_x(I) = \min\{\mathrm{wt}(u) : u \in I^\perp,\ u_x \neq 0\} + d(I^\perp).$$

For any fixed radius r and any L-linear outer family whose dual distance tends to infinity, all sufficiently long concatenations confine every radius-r recovery equation at a designated copy of x to its inner block if and only if $r + 1 < z_x(I)$. Whenever this inequality holds, the complete family of exact helper supports and all normalized dual words are copied exactly in every block of each such sufficiently long concatenation. Choosing an asymptotically good outer family therefore realizes the prescribed bounded recovery structure on a coordinate class of density $1/|E|$ in an asymptotically good fixed-alphabet family.

Moreover, over one finite field there are two represented $[10,4,6]$ inner codes with the same pointed rank-triple multiplicity enumerator, locality three, five minimum repairs, matching number two, transversal number two, a unique minimum blocker, and the same helper degree multiset. For one common outer family, matched length and dimension formulas and the same distance lower bound coexist with density-$1/10$ target classes carrying the two distinct reliability laws displayed in the abstract.

The first paragraph is the exact confinement criterion of Theorems 3.1 and 4.1; the second is Theorem 6.5. The obstruction $z_x(I)$ is the least pointed inner-dual word in the target block plus the least nonzero inner-dual word in another block. Nonzero outer functionals eventually become too expensive, but this all-zero-functional obstruction persists.

Theorem 1.2 (MDS reconstruction from minimum-weight recovery equations)**.** *Let $C \le \mathbb{F}_q^E$ be an $[n,k]_q$ MDS code with $1 \le k < n$, and let $x \in E$. Then*

$$\operatorname{span} \mathcal{P}_x^{(\le k)}(C) = C^\perp, \qquad \rho_x(C) = k,$$

where $\rho_x(C)$ is the least reconstruction radius. The supports of the minimum-weight recovery equations form the complete k-uniform clutter on $E \setminus \{x\}$.

The supports are generic for these MDS parameters, but the normalized equations span the represented dual code. Minimum dual words with a common core of $k-1$ helpers and distinct private coordinates form a basis, so their coefficients retain information that support-only locality discards.

The main separation matches the pointed rank-triple enumerator, repair count, locality, matching and transversal numbers, minimum-blocker count, and helper degrees of two represented sparse-paving seeds. Higher circuit intersections still produce different reliability and bounded-EXIT curves. A common outer family transfers the difference to target classes of density $1/10$ with matched global bounds. A quotient-plane line arrangement and generic lift give the structural seed proof.

Figure 1 gives the transfer and reliability chain.

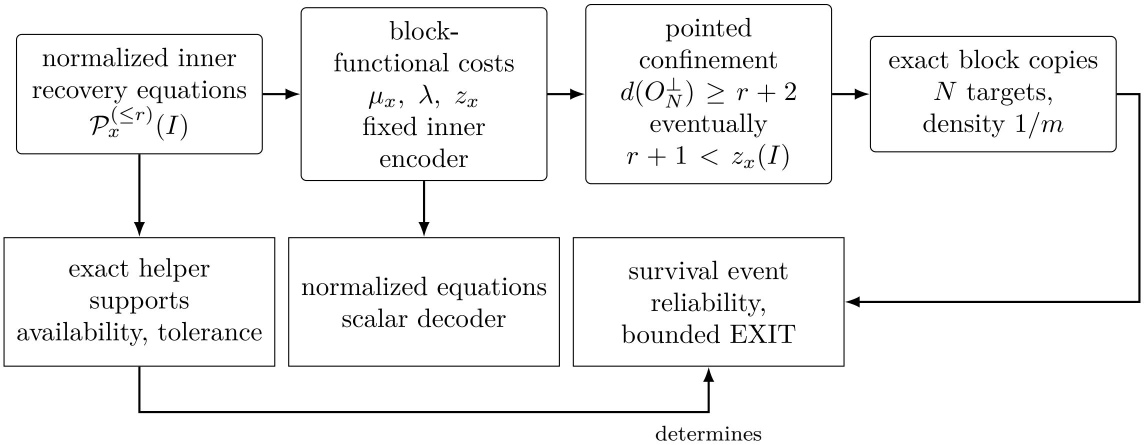

Figure 1: The transfer spine. In the fixed-inner, L-linear concatenation regime, weighted block-functional costs give an exact confinement gate. When $d(O_N^\perp) \to \infty$ and $r+1 < z_x(I)$, exact helper supports and normalized equations are copied into every block. Reliability is derived from the copied support clutter; the diagram makes no necessity claim for arbitrary asymptotically good families.

The asymptotic coding ingredients, reliability identities, and pointed-Tutte polynomial are classical. We organize them around bounded pointed recovery, the transfer boundary, positive-density realization, and two geometric recovery structures. We make no minimum-bandwidth claim, no full-MAP claim for a radius cutoff, and no threshold claim for harmonic closure. In the literature reviewed, we did not locate the exact bounded-recovery confinement criterion or the two coordinatewise recovery-structure inventories [15, 11, 8, 5]; this is a bounded comparison, not a priority claim.

2 Bounded recovery structures

Let $C \le \mathbb{F}_q^X$ be a linear code, let $x \in X$, and let $r \ge 0$.

Throughout, whenever matching, transversal, failure tolerance $\tau - 1$, or a minimum-blocker leading term is asserted, the minimal repair clutter is nonempty and does not contain the empty helper set. The support, coefficient, and reliability definitions themselves remain valid in the two degenerate cases: an empty support family never repairs, while an empty repair edge repairs without a helper. We state those cases separately rather than assign them an adversarial-tolerance interpretation. We use the extended-distance convention $d(\{0\}) = \infty$.

Definition 2.1 (Exact helper-support layer)**.** The complete radius-r exact-support hypergraph at x is

$$\mathcal{H}_x^{(\le r)}(C) = \Big\{ R \subseteq X \setminus \{x\} : |R| \le r,\ \exists w \in C^\perp,\ w_x \ne 0,\ \mathrm{supp}(w) \setminus \{x\} = R \Big\}.$$

Its minimal members form the repair clutter $\min \mathcal{H}_x^{(\le r)}(C)$. Write $\nu_x^{(\le r)}$ and $\tau_x^{(\le r)}$ for its matching and transversal numbers.

The distinction between the complete hypergraph and its minimal clutter is useful: the former records every bounded witness, while every monotone support invariant below depends only on the latter.

Proposition 2.2 (Basic invariants)**.** *Assume the empty helper set is not a repair. Then*

$$\nu_x^{(\leq r)} \leq \tau_x^{(\leq r)}.$$

Both numbers are unchanged after deleting nonminimal repair sets, so either the complete hypergraph or its minimal clutter may be used for these optima.

Proof. A matching of m nonempty edges forces every transversal to select at least one point from each edge. Minimalization preserves the upward closure of the success event, hence preserves matching and transversal optima. □

Every member of $\mathcal{H}_x^{(\leq r)}(C)$ is a recovery set in the standard sense. The standard bounded recovery-set family is its upward closure within the helper bound: it consists of the sets containing some member of $\mathcal{H}_x^{(\leq r)}(C)$. Thus $\mathcal{H}_x^{(\leq r)}(C)$ records the helpers with nonzero coefficients, while the upward closure records all sets from which recovery is possible.

Definition 2.3 (Normalized recovery equations)**.** Write

$$\mathcal{P}_x^{(\leq r)}(C) = \{w \in C^\perp : w_x = 1,\ |\operatorname{supp}(w) \setminus \{x\}| \leq r\}$$

for the normalized bounded recovery equations through x. For $R \in \mathcal{H}_x^{(\leq r)}(C)$, its coefficient fiber is the set of normalized coefficient vectors

$$\left(-\frac{w_y}{w_x}\right)_{y\in R} \qquad (w \in C^\perp,\ \operatorname{supp}(w) = R \cup \{x\}).$$

Each vector gives the exact recovery equation $c_x = \sum_{y\in R}(-w_y/w_x)c_y$. Operationally, a helper transmits its stored scalar in $\mathbb{F}_q$ and the decoder applies the displayed coefficient. Thus a radius-r repair contacts at most r helpers and downloads one whole base-field symbol from each. This scalar exact-repair model does not optimize subsymbol access or repair bandwidth in an array-code model [9, 23, 17].

Definition 2.4 (Reconstruction radius)**.** These equations *reconstruct* C at radius r when their linear span is $C^\perp$; equivalently, double duality recovers C from them. Its reconstruction radius is the least such r, or infinity if no such radius exists.

Proof of Theorem 1.2. The dual has dimension $n-k$ and minimum distance $k+1$. Given any $(k+1)$-set $T \subseteq E$, restrict $C^\perp$ to $E \setminus T$. Its domain has dimension $n-k$ and its codomain dimension $n-k-1$, so the restriction has a nonzero kernel vector y supported on T. The dual-distance bound forces $\operatorname{supp}(y) = T$. If $x \in T$, then $y_x \neq 0$ and scaling gives a normalized repair word. Thus every k-subset of $E \setminus \{x\}$ is a minimum helper set. Conversely, any normalized repair with at most k helpers has total weight at most $k+1$, so the same distance bound forces equality. This proves the support-clutter assertion.

Choose $Q \subseteq E \setminus \{x\}$ with $|Q| = k-1$ and set $U = E \setminus (\{x\} \cup Q)$. For every $t \in U$, choose the normalized minimum dual word y_t supported on $\{x,t\} \cup Q$. The coordinate t is nonzero in y_t and zero in every y_s with $s \neq t$. Evaluating a linear relation at the private coordinates t shows that the y_t are independent. There are $|U| = n-k = \dim C^\perp$ of them, so they form a basis of $C^\perp$ contained in the radius-k equation family. Double duality therefore recovers C. No smaller radius contains any normalized word, since that would produce a nonzero dual word of weight at most k. Hence $\rho_x(C) = k$. □

Definition 2.5 (Probability layer)**.** Let $A \subseteq X \setminus \{x\}$ be the surviving helpers and put

$$f_x^{(\le r)}(A) = 1 \quad \iff \quad \exists R \in \mathcal{H}_x^{(\le r)}(C) \text{ with } R \subseteq A.$$

For independent survival probabilities (s_y), the reliability is $R_x^{(\le r)}(s) = \mathbb{E}[f_x^{(\le r)}]$.

Exact equality of the helper supports and equations preserves all support-derived invariants: matching, transversals, blocker clutters, multivariate reliability, and every correlated-law reliability after transporting the helper labels. Coefficient fibers require the finer functional data used next.

3 Exact weighted-functional transfer

Let $I \le \mathbb{F}_q^m$ be an inner $[m,k,d]_q$ code and identify its message space with $L = \mathbb{F}_{q^k}$ through an $\mathbb{F}_q$-linear encoder $e : L \to I$. For $w \in \mathbb{F}_q^m$, define the induced block functional

$$\Phi_I(w) : L \longrightarrow \mathbb{F}_q, \qquad a \longmapsto \langle w, e(a) \rangle.$$

The zero fiber is $I^\perp$. For $\beta \in \mathrm{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$ put

$$\lambda(\beta) = \min_{\Phi_I(w)=\beta} \mathrm{wt}(w), \qquad \mu_x(\beta) = \min_{\substack{\Phi_I(w)=\beta \\ w_x \ne 0}} \mathrm{wt}(w),$$

with value ∞ for an empty constrained fiber.

Let $O \le L^J$ be an outer code. Its functional dual consists of tuples $(\beta_j)_{j\in J}$ satisfying

$$\sum_{j \in J} \beta_j(a_j) = 0 \qquad \text{for every } a = (a_j) \in O.$$

The weighted cost of such a tuple is $\sum_j \lambda(\beta_j)$. This is the ordinary quotient/coset-leader weight attached to the inner encoder; the pointed use below is the repair-specific part. Write

$$\mathrm{fwt}(\beta) = \big|\{j : \beta_j \ne 0\}\big|.$$

Thus the functional-dual tuples have three disjoint strata: $\mathrm{fwt}(\beta) = 0$, $\mathrm{fwt}(\beta) = 1$, and $\mathrm{fwt}(\beta) \ge 2$. Keeping the singleton stratum is essential when an outer coordinate projection is not surjective. If every coordinate projection of O is surjective, a functional-dual tuple cannot have functional weight one.

The all-zero functional sector has the fixed pointed cost

$$z_x(I) = \begin{cases} \mu_x(0) + d(I^\perp), & I^\perp \ne 0, \\ \infty, & I^\perp = 0. \end{cases}$$

This is the persistent confinement threshold: it is the least total weight of a nonzero inner-dual word through the target together with a nonzero inner-dual word in another block. For a fixed outer code, the nonzero functional sector below remains part of the exact confinement gate.

Theorem 3.1 (Exact pointed confinement and transfer)**.** *Assume $|J| \ge 2$, and fix an outer block j and an inner coordinate x. The minimum weight of a concatenated dual witness through (j,x) that is not an embedded inner-dual witness is*

$$\min \left\{ z_x(I), \min_{\substack{(\beta_\ell) \text{ in the functional dual of } O \\ (\beta_\ell) \ne 0}} \left(\mu_x(\beta_j) + \sum_{\ell \ne j} \lambda(\beta_\ell) \right) \right\},$$

where a minimum over an empty set is ∞. Consequently, if this pointed cost is greater than $r+1$, then

$$\mathcal{H}^{(\le r)}_{(j,x)}(O \circ I) = \mathrm{embed}_j\big(\mathcal{H}^{(\le r)}_x(I)\big).$$

Under the same hypothesis, zero-extension is a bijection

$$\mathcal{P}^{(\le r)}_x(I) \xrightarrow{\sim} \mathcal{P}^{(\le r)}_{(j,x)}(O \circ I).$$

In particular, with the convention $d(\{0\}) = \infty$, it suffices that

$$r + 1 < 2d(I^{\perp}) \quad \textit{and} \quad \sum_j \lambda(\beta_j) \ge r + 2$$

for every nonzero functional-dual tuple.

Proof. Let $w = (w_\ell)_{\ell \in J}$ be a block word and set $\beta_\ell = \Phi_I(w_\ell)$. For every outer word $a = (a_\ell) \in O$,

$$\langle w, e(a)\rangle = \sum_{\ell \in J} \langle w_\ell, e(a_\ell)\rangle = \sum_{\ell \in J} \beta_\ell(a_\ell).$$

Hence $w \in (O \circ I)^{\perp}$ if and only if $\beta = (\beta_\ell)$ belongs to the functional dual of O. This calculation is the only concatenated-dual decomposition used below; no fiber enumerator is needed.

Fix such a tuple β. The constraints on the blocks are independent. At the target, the condition $w_j(x) \ne 0$ gives minimum $\mu_x(\beta_j)$. At every other block the minimum is $\lambda(\beta_\ell)$. Each ordinary fiber is nonempty because a functional on the encoded inner space extends to the ambient coordinate space, and its minimum is attained because Hamming weight takes values in the well-ordered set $\mathbb{N}$. The constrained target fiber is either empty, in which case its cost and the fixed-fiber cost are infinite, or has an attained minimum by the same argument. Therefore the exact pointed minimum in the fixed fiber β is

$$\mu_x(\beta_j) + \sum_{\ell \ne j} \lambda(\beta_\ell).$$

If $\beta \ne 0$, a realization cannot be an embedded inner-dual word in block j, since every embedded inner-dual word induces the zero functional in every block. Taking the minimum over the nonzero functional dual gives the second term in the theorem. This minimum includes both remaining strata: functional weight one and functional weight at least two.

It remains to analyze $\beta = 0$. Then every w_ℓ lies in $I^{\perp}$. A pointed realization that is not embedded in block j consists of a word in $I^{\perp}$ that is nonzero at x, together with a nonzero word of $I^{\perp}$ in at least one off-target block. Conversely, placing any such pair in two blocks and putting zero elsewhere gives a pointed nonembedded concatenated-dual word. Since $|J| \ge 2$, the two choices can be made independently, and their exact minimum is $\mu_x(0) + d(I^{\perp}) = z_x(I)$. If either required word does not exist, both the constrained minimum and $z_x(I)$ are infinite. The zero and nonzero functional sectors partition all pointed nonembedded witnesses, proving the displayed minimum formula.

Suppose that this minimum is greater than $r+1$, and take a radius-r pointed repair witness in the concatenated code. Its total weight is at most $r+1$, so it cannot be nonembedded. It is therefore the zero-extension of an inner-dual word in block j. Deleting the target coordinate identifies its helper support with the image under embed_j of a radius-r inner helper support. This proves one inclusion of the asserted exact-support equality. If the witness is normalized at the target, its nonzero block is the correspondingly normalized inner word, so this identification also gives the asserted bijection of normalized dual words. Conversely, the zero-extension of every radius-r inner repair word is orthogonal to the concatenated code, and its helper support is the corresponding embedded support; normalization is unchanged. This proves the reverse inclusions in both layers.

Finally, whenever the zero-sector cost is finite, $\mu_x(0) \geq d(I^\perp)$, so $z_x(I) \geq 2d(I^\perp)$. For a nonzero functional tuple, $\mu_x(\beta_j) \geq \lambda(\beta_j)$, and hence its pointed cost is at least $\sum_\ell \lambda(\beta_\ell)$. The two displayed numerical hypotheses therefore put every pointed nonembedded witness above weight $r+1$, and the transfer conclusion follows. ☐

The functional-dual decomposition is classical concatenation machinery [2, Theorem 2.3][3, Theorem 2.1]. The weighted statement matters because ordinary functional support distance can be strictly weaker.

Corollary 3.2 (Strict weighted transfer)**.** *Let I be the completed $[20,4,9]_9$ cubic–axis code of Section 7, identify its message space with $L = \mathbb{F}_{9^4}$, and let $a : L \to L$ be an $\mathbb{F}_9$-linear Singer automorphism chosen as in the proof below. The generalized single-parity-check outer code*

$$O_a = \{u \in L^5 : u_0 + a(u_1) + u_2 + u_3 + u_4 = 0\}$$

has functional-dual support distance five, but its nonzero weighted realization cost for I is at least six. Thus the ordinary support-distance condition at radius four fails while the exact weighted gate holds, and every radius-four exact support and equation family transfers to each of the five blocks of $O_a \circ I$.

Proof. The nonzero projective functionals on the four-dimensional $\mathbb{F}_9$-space L form $(9^4-1)/(9-1) = 820$ classes. Let S be the set of the twenty coordinate-functional classes of I; they are distinct because $d(I^\perp) = 3$. Write $a^*[f] = [f \circ a]$ for the induced Singer action. Regularity [18] supplies a with $a^*(S) \cap S = \varnothing$: at most $20^2 = 400 < 820$ group elements map some selected class to another selected class.

Every functional-dual tuple of O_a is $(f, f \circ a, f, f, f)$. If $f \neq 0$, all five entries are nonzero, so the functional support distance is exactly five. A weight-five realization would have unit cost in every block. In particular, $[f] \in S$ and $a^*[f] = [f \circ a] \in S$, so $a^*[f] \in a^*(S) \cap S$, a contradiction. Hence every nonzero tuple costs at least six. The zero sector also costs at least $2d(I^\perp) = 6$, so Theorem 3.1 with $r = 4$ gives the two exact recovery-data equalities. The displayed parity check makes every outer coordinate projection surjective. Taking any nonzero f shows at the same time that functional distance six is false. ☐

4 Prescribed positive-density realization

The pointed zero-functional sector persists even when outer dual distance grows.

Theorem 4.1 (Prescribed represented recovery structures)**.** *Let $O_N \leq L^N$ be L-linear outer codes with $d(O_N^\perp) \to \infty$, and let $C_N = O_N \circ I$. For all sufficiently large N, every pointed dual witness through (j,x) of weight at most $r+1$ is confined to block j if and only if*

$$r + 1 < z_x(I).$$

Under this condition the radius-r helper supports and normalized equations at x occur on the designated class of N targets (j,x), a class of density exactly $1/m$ in C_N. Other inner coordinate types may carry isomorphic recovery data as well. More precisely, at every block

$$\mathcal{H}^{(\leq r)}_{(j,x)}(C_N) = \mathrm{embed}_j\big(\mathcal{H}^{(\leq r)}_x(I)\big), \qquad \mathcal{P}^{(\leq r)}_{(j,x)}(C_N) = \mathrm{embed}_j\big(\mathcal{P}^{(\leq r)}_x(I)\big).$$

Proof. Every $\mathbb{F}_q$-linear functional on L has a unique form $a \mapsto \mathrm{Tr}_{L/\mathbb{F}_q}(ba)$, by nondegeneracy of the trace pairing. Let $\beta = (\beta_j)$ lie in the functional dual of the outer code, and write $\beta_j(a) = \mathrm{Tr}(b_j a)$. Since O_N is L-linear, applying β to every scalar multiple ca of every $a \in O_N$ shows

$$\mathrm{Tr}\left(c \sum_j b_j a_j\right) = 0 \quad \text{for every } c \in L.$$

Trace nondegeneracy gives $(b_j) \in O_N^\perp$. Conversely, every outer dual word gives such a functional tuple. The trace representation is coordinatewise injective, so it preserves support exactly. Thus the functional-dual support distance is $d(O_N^\perp)$.

Fix r. For all sufficiently large N, $d(O_N^\perp) \geq r+2$. Every realization of a nonzero functional tuple then has weight at least $r+2$: each nonzero functional requires a nonzero inner block. The exact pointed decomposition of Theorem 3.1 consequently leaves only the zero-functional sector in the bounded range. For $N \geq 2$, its exact cost is the block-independent quantity $z_x(I)$. Hence confinement through every (j, x) holds exactly when $r + 1 < z_x(I)$.

If $r + 1 < z_x(I)$, Theorem 3.1 identifies both exact helper supports and normalized equations by zero-extension. If $r+1 \geq z_x(I)$, the attained pointed inner-dual minimum at x, together with a minimum nonzero inner-dual word in a second block, gives a zero-functional pointed nonembedded witness of weight at most $r + 1$. This proves necessity as well as sufficiency. Finally, the N designated targets (j, x) among the mN concatenated coordinates form a class of density exactly $1/m$, irrespective of any additional occurrences. □

Proposition 4.2 (Concatenated parameters)**.** *If I has parameters $[m, k, d]_q$ and $O_N \leq L^N$ has L-dimension K_N and symbol distance D_N, then $C_N = O_N \circ I$ has length mN, $\mathbb{F}_q$-dimension kK_N, and minimum distance at least dD_N. In particular, if*

$$\frac{K_N}{N} \longrightarrow R > 0, \qquad \liminf_N \frac{D_N}{N} \geq \delta > 0,$$

then the concatenated family has asymptotic rate kR/m and relative distance at least $d\delta/m$.

Proof. The blockwise inner encoder is injective, so restriction of scalars multiplies the outer L-dimension by $[L : \mathbb{F}_q] = k$. A nonzero outer word has at least D_N nonzero symbols, and each nonzero symbol encodes to an inner word of weight at least d. The finite parameter statements follow; division by mN gives the asymptotic claims. □

Corollary 4.3 (Positive-density MDS coefficient fingerprints)**.** *Let I be an $[m, k]_q$ MDS code with $1 \leq k < m$, and fix x. Its radius-k exact helper-support family is the complete k-uniform clutter on the other coordinates, while its minimum-weight normalized recovery equations reconstruct I, and $z_x(I) = 2(k+1)$. Both layers occur on a designated target class of density $1/m$ in every sufficiently long concatenation by an outer family with dual distance tending to infinity.*

Proof. The minimum-weight assertions are Theorem 1.2. Every minimum dual support through x has size $k+1$, so $\mu_x(0) = d(I^\perp) = k+1$ and

$$z_x(I) = 2(k+1) > k+1.$$

Apply Theorem 4.1 with $r = k$. The helper-support family depends only on the MDS parameters; the transferred normalized coefficients retain the represented inner code. □

Corollary 4.4 (Positive-density Clebsch recovery equations)**.** *Let I be the $[6,3,4]_{11}$ Clebsch hexagon code [16], and fix a coordinate x. The minimal radius-three repair clutter at x is the complete three-uniform hypergraph on the other five coordinates. Hence it has ten minimal repair supports,*

$$\nu_x^{(\leq 3)} = 1, \qquad \tau_x^{(\leq 3)} = 3,$$

and homogeneous helper-survival reliability

$$10s^3(1-s)^2 + 5s^4(1-s) + s^5.$$

Already the radius-three equations at x determine I. Moreover $z_x(I) = 8$, so both the minimum-weight radius-three recovery equations and the full radius-five equation family occur on a designated target class of density $1/6$ in an asymptotically good fixed-$\mathbb{F}_{11}$ family.

Proof. The dual $I^\perp$ is again an $[6,3,4]_{11}$ MDS code. Every four-set containing x is therefore the support of a dual word, unique up to scalar. The minimal helper sets are consequently all $\binom{5}{3} = 10$ triples on the other coordinates. Their matching and transversal numbers are one and three, and recovery succeeds exactly when at least three of the five helpers survive, giving the displayed polynomial.

At radius three the common-core star of minimum normalized dual words is already a basis of $I^\perp$, by Corollary 4.3. Thus the minimum-weight normalized recovery equations, and hence also the full radius-five family, recover $I^\perp$ and I.

Finally, the minimum dual-word weight through x is four, so

$$z_x(I) = \mu_x(0) + d(I^\perp) = 4 + 4 = 8.$$

Taking $r = 5$ gives $r + 1 = 6 < 8$. Theorem 4.1 therefore confines every full pointed witness to its inner block and reproduces the coefficient fibers at one of the six coordinate types in every block. Choosing asymptotically good outer codes over $\mathbb{F}_{11^3}$ gives the claimed fixed-$\mathbb{F}_{11}$ family and density. □

Remark 4.5. The support clutter and reliability polynomial in Corollary 4.4 are shared by every $[6,3,4]$ MDS code. The Clebsch-specific information lies in the scalar coefficient fibers: they recover the inner code that is characterized geometrically in [16].

Remark 4.6. Corollary 4.3 applies without a new transfer argument to every represented MDS inner code. This includes the usual arc and projective Reed–Solomon presentations, and the classical MDS code presentations underlying AME constructions, whenever their displayed field and dimension hypotheses hold. Their minimum exact helper-support families are generic MDS data; only their normalized equations distinguish the represented geometric or quantum-code realization.

This is a represented-recovery theorem: it does not say that every abstract hypergraph is linearly representable. It yields standard scaled asymptotic regions. If $Q = q^k$, random L-linear outer codes of rate R can have relative distances $\delta, \delta^\perp$ whenever

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^\perp) < R.$$

For an inner $[m,k,d]_q$ code the concatenated parameters satisfy

$$R_{\text{concat}} \to \frac{k}{m} R, \qquad \delta_{\text{concat}} \geq \frac{d}{m} \delta.$$

Indeed, for a random L-linear outer code the expected numbers of nonzero words and dual words below the two cutoffs have exponential rates $R + H_Q(\delta) - 1$ and $H_Q(\delta^\perp) - R$, respectively, so both vanish simultaneously; this is the random-linear-code form of the Gilbert–Varshamov argument [12, Chapter 17]. When Q is square, algebraic-geometry outer codes similarly give, for $\gamma = 1/A(Q)$ and $\gamma < R < 1 - \gamma$,

$$\delta(O_N) \geq 1 - R - \gamma, \qquad \delta(O_N^\perp) \geq R - \gamma, \qquad \delta_{\text{concat}} \geq \frac{d}{m}(1 - R - \gamma).$$

These are the classical GV and TVZ ingredients. For the completed $[20, 4, 9]_9$ seed, $Q = 6561$ and $\gamma \leq 1/80$, recovering the concrete rate-1/10 family as one instance [20, Chapters 2 and 7]. The stronger self-dual family used for that concrete instance is precisely Stichtenoth's theorem over $\mathbb{F}_{6561}$ [19, Theorem 1.6(ii)].

5 Reliability and bounded EXIT

Let $\mathcal{H}$ be a repair clutter on helper set V. For $v \in V$, define

$$\mathcal{H} - v = \min\{E \in \mathcal{H} : v \notin E\}, \qquad \mathcal{H}/v = \min\{E \setminus \{v\} : E \in \mathcal{H}\}.$$

Theorem 5.1 (Reliability calculus)**.** *For independent helper survival probabilities (s_v),*

$$R_{\mathcal{H}}(s) = (1 - s_v)R_{\mathcal{H}-v}(s) + s_v R_{\mathcal{H}/v}(s),$$

and

$$\frac{\partial R_{\mathcal{H}}}{\partial s_v} = R_{\mathcal{H}/v} - R_{\mathcal{H}-v} = \Pr(v \textit{ is pivotal}).$$

Under homogeneous survival s, $R'_{\mathcal{H}}(s) = \sum_v \Pr_s(v$ is pivotal$)$. If the minimum blocker size is τ and there are b_τ minimum blockers, then for helper failure probability p,

$$1 - R_{\mathcal{H}}(1 - p) = b_\tau p^\tau + O(p^{\tau+1}).$$

Proof. Write $f_{\mathcal{H}}(A) = 1$ when A contains an edge of $\mathcal{H}$, and write it as zero otherwise. The reliability is the finite multilinear sum

$$R_{\mathcal{H}}(s) = \sum_{A \subseteq V} f_{\mathcal{H}}(A) \prod_{y \in A} s_y \prod_{y \notin A} (1 - s_y).$$

Partition the subsets according to whether they contain v. Erasing v from those that do contain it identifies the two parts with the Boolean systems $\mathcal{H} - v$ and $\mathcal{H}/v$, respectively. This gives

$$R_{\mathcal{H}} = (1 - s_v)R_{\mathcal{H}-v} + s_v R_{\mathcal{H}/v}.$$

The two conditional reliabilities do not involve s_v, so differentiation gives $R_{\mathcal{H}/v} - R_{\mathcal{H}-v}$. Monotonicity makes this difference the conditional probability that adjoining v changes failure into success, which is precisely pivotality. On the diagonal $s_v = s$, the chain rule sums these coordinate derivatives and gives the Russo–Margulis identity.

Now expose the failed set F. Repair fails exactly when F is a transversal of $\mathcal{H}$, equivalently when it contains a minimal blocker. Thus

$$1 - R_{\mathcal{H}}(1 - p) = \sum_{\substack{F \subseteq V \\ F \text{ is a transversal}}} p^{|F|}(1 - p)^{|V| - |F|}.$$

There are no summands below degree τ, and the transversals of size τ are exactly the b_τ minimum blockers. Their contribution is $b_\tau p^\tau (1-p)^{|V|-\tau} = b_\tau p^\tau + O(p^{\tau+1})$. Every remaining term contains at least $p^{\tau+1}$; the sum is finite, proving the expansion. □

These are the standard deletion–contraction and pivotality identities for a monotone reliability system [4].

For EXIT conventions, condition the target x unavailable and erase helper v with probability p_v. Put

$$h_x^{(\le r)}(p) = \Pr(f_x^{(\le r)}(A) = 0), \qquad R_x^{(\le r)} = 1 - h_x^{(\le r)}.$$

This is an *extrinsic* failure probability. If the target is itself erased with probability p_x, its residual erasure probability is $p_x h_x^{(\le r)}$. At full radius it is symbol-MAP EXIT; at finite radius it is a bounded-query decoder and carries no capacity claim [1, 14].

Proposition 5.2 (Bounded-EXIT hierarchy)**.** *The failure form of deletion–contraction is*

$$h_{\mathcal{H}}(p) = p_v h_{\mathcal{H}-v}(p) + (1-p_v) h_{\mathcal{H}/v}(p), \qquad \frac{\partial h_{\mathcal{H}}}{\partial p_v} = \Pr(v \textit{ is pivotal}).$$

If $L_x(A)$ is the size of the cheapest repair contained in the realized survivor set, with $L_x = \infty$ when none exists, then

$$\Pr(L_x = r) = h_x^{(\le r-1)} - h_x^{(\le r)}, \qquad \Pr(L_x = \infty) = h_x^{\mathrm{MAP}}.$$

The first identity is for $r \ge 1$.

Proof. Conditioning on whether v is erased gives the displayed recurrence: when it is erased, only repairs avoiding v remain; when it survives, it may be contracted from every repair. Equivalently, substitute $s_y = 1 - p_y$ in the reliability recurrence and take complements. Differentiation reverses both signs, so the derivative is again the pivotal probability.

Let $\mathcal{H}_x^{(\le r)}$ be the full exact-support family filtered by edge size at most r. The event $L_x \le r$ is exactly success for this bounded family. Hence

$$\{L_x = r\} = \{L_x > r-1\} \setminus \{L_x > r\},$$

and taking probabilities gives the difference of failure curves. At full radius, the complementary event is that no repair exists, which is $L_x = \infty$. □

Minimal transversals of $\mathcal{H}_x^{(\le r)}$ are exact target-specific, one-shot bounded-repair failure certificates. They are not automatically the representation-dependent stopping sets of a global iterative Tanner decoder.

For a length-N code define the radius-r locality deficit

$$L_r(x) = \int_0^1 \left(h_x^{(\le r)}(p) - h_x^{\mathrm{MAP}}(p) \right) dp.$$

If Δa_k counts size-k survivor sets repaired by full MAP but not at radius r, beta integration gives

$$L_r(x) = \sum_{k=0}^{N-1} \frac{\Delta a_k}{N\binom{N-1}{k}}.$$

With entropy normalized in $\log_q$ units, the unrecoverable-symbol indicator is the normalized conditional entropy on the q-ary erasure channel. Under this convention the symbol-MAP areas sum to the code dimension K [1], so

$$\sum_x \int_0^1 h_x^{(\le r)}(p)\, dp = K + \sum_x L_r(x).$$

6 Pointed Tutte structure

Let M be the column matroid of a projective-system code on $E = V \sqcup \{x\}$, with x neither a loop nor a coloop. Put

$$\epsilon_x(A) = r_M(A \cup \{x\}) - r_M(A) \in \{0, 1\}.$$

The set A repairs x exactly when $\epsilon_x(A) = 0$.

Theorem 6.1 (Pointed perspective polynomial)**.** *Let $D = M\backslash x$ and $Q = M/x$. The Las Vergnas polynomial of the elementary perspective $D \to Q$ is*

$$T_{D\to Q}(X, Y, Z) = \sum_{A\subseteq V} (X - 1)^{r(M)-r_M(A\cup\{x\})}(Y - 1)^{|A|-r_M(A)} Z^{\epsilon_x(A)}.$$

If

$$S_x(u) = \sum_{\epsilon_x(A)=0} u^{|A|},$$

then, for $n = |V|$,

$$R_{(M,x)}(s) = \sum_{\epsilon_x(A)=0} s^{|A|}(1 - s)^{n-|A|}.$$

Consequently, when $s \neq 1$,

$$R_{(M,x)}(s) = (1 - s)^n S_x\left(\frac{s}{1 - s}\right).$$

Writing $U_K(u, y) = \sum_A u^{|A|}y^{r_K(A)}$, one also has

$$S_x(u) = \frac{\partial}{\partial y}\big(U_D(u, y) - U_Q(u, y)\big)\bigg|_{y=1}.$$

Proof. Contraction gives $r_D(A) = r_M(A)$ and $r_Q(A) = r_M(A \cup \{x\}) - 1$, so $r_D(A) - r_Q(A) = 1 - \epsilon_x(A)$. Substitution in the rank-one perspective expansion gives the displayed polynomial term by term. The Z^0 terms are exactly the successful sets; more explicitly, for $u \neq 0$,

$$S_x(u) = u^{r(M)}[Z^0]T_{D\to Q}(1 + u^{-1}, 1 + u, Z),$$

because a term with exponents $a, b, 0$ has $|A| = r(M) - a + b$. Multiplying the coefficient of $u^{|A|}$ by $s^{|A|}(1 - s)^{n-|A|}$ and summing gives the reliability formula. Finally,

$$\partial_y(y^{r_D(A)} - y^{r_Q(A)})\Big|_{y=1} = r_D(A) - r_Q(A),$$

which is one when $\epsilon_x(A) = 0$ and zero when $\epsilon_x(A) = 1$. Summing the differentiated terms proves the last identity. □

This identification is the singleton specialization of standard set-pointed Tutte theory [22, equation (3.1)]; it is not a new polynomial.

Corollary 6.2 (Pointed repair–failure duality)**.** *Pointed duality gives*

$$R_{(M,x)}(s) + R_{(M^*,x)}(1 - s) = 1,$$

and the minimal failure blockers of (M, x) are exactly the minimal repair sets of (M^*, x). For a represented matroid they are the sets $Q' \setminus \{x\}$, where Q' ranges over cocircuits containing x; equivalently, Q' is an inclusion-minimal row-code support through x. In particular,

$$\tau_{\text{full}}(x) = d_x(C) - 1, \qquad d_x(C) = \min\{\text{wt}(c) : c \in C,\ c_x \neq 0\}.$$

Proof. For $F \subseteq V$, put $A = V \setminus F$. The dual rank formula, applied once to F and once to $F \cup \{x\}$, gives

$$r_{M^*}(F \cup \{x\}) - r_{M^*}(F) = 1 - \epsilon_x(A).$$

Thus F repairs x in M^* exactly when its complement fails to repair x in M. Complementation preserves the product measure after replacing s by $1-s$, which proves the reliability identity. It also exchanges minimal dual repairs with minimal primal failure blockers. Circuits of M^* are cocircuits of M, so the blockers are $Q' \setminus \{x\}$ for cocircuits Q' through x. In a represented matroid, cocircuits are the inclusion-minimal nonzero row-code supports. Taking the smallest support through x proves the distance formula. □

Lemma 6.3 (Sparse-paving pointed enumerator)**.** *For a rank-r sparse paving matroid on n elements with distinguished element x, the multiplicity enumerator of the rank triples $(r_M(A), r_M(A \cup \{x\}), |A|)$, over all $A \subseteq E \setminus \{x\}$, is determined by n, r, the number of circuit-hyperplanes containing x, and the number avoiding x. Equivalently, these data determine the full pointed-perspective polynomial. They do not determine the labeled subset-rank function $A \mapsto r_M(A)$.*

Proof. Every set of size below r is independent, every set of size above r has full rank, and an r-set has rank $r-1$ exactly when it is a circuit-hyperplane. Hence the rank triple

$$\bigl(r_M(A), r_M(A \cup \{x\}), |A|\bigr), \qquad A \subseteq E \setminus \{x\},$$

has its uniform-matroid value except when A or $A \cup \{x\}$ is one of the two indicated types of circuit-hyperplane. Their two counts therefore determine the multiplicity of every rank triple. □

Proposition 6.4 (The radius filtration is not a pointed-Tutte specialization)**.** *The complete pointed polynomial does not determine the bounded-radius reliability filtration, even among rank-four systems represented over $\mathbb{F}_7$.*

Proof. The mechanism is sparse paving. Lemma 6.3 does not record how the circuit-hyperplanes through x meet after x is removed. The radius-$(r-1)$ reliability does record those intersections, because it is the probability of the union of their containment events.

Distinguish column 0 in each of the following matrices; columns $1, \dots, 6$ are the helpers:

$$G_{\mathrm{dis}} = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 0 & 3 & 1 & 4 & 0 & 6 & 1 \\ 0 & 2 & 3 & 2 & 1 & 5 & 0 \\ 0 & 1 & 4 & 1 & 2 & 4 & 3 \end{pmatrix}, \qquad G_{\mathrm{ov}} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 5 & 4 & 0 & 6 \\ 0 & 4 & 2 & 0 & 2 & 1 & 2 \\ 0 & 4 & 1 & 6 & 4 & 5 & 4 \end{pmatrix}.$$

All entries and determinants below are in $\mathbb{F}_7$. Every three-column set is independent. The dependent four-column sets are, respectively,

$$\{0145, 0236, 2345\}, \qquad \{0123, 0256, 3456\}.$$

Here, for example, 0145 denotes $\{0, 1, 4, 5\}$. Direct elimination gives the two lists: all other four-column minors are nonzero, and every triple extends to one of them. The complete determinant ledger and exact replay are recorded in Appendix A.1. Thus each displayed representation has rank four, has no circuit of size at most three, and has exactly the three listed circuit-hyperplanes.

Both matroids are rank-four sparse paving matroids with three circuit-hyperplanes, two through the distinguished column and one avoiding it. The preceding sparse-paving count therefore gives, for both systems,

$$\begin{aligned} T(X, Y, Z) = {} & (X-1)^3 Z + 6(X-1)^2 Z + 15(X-1)Z + 2(X-1) \\ & + 18Z + (Y-1)Z + 14 + 6(Y-1) + (Y-1)^2. \end{aligned}$$

Equivalently, their complete successful-set enumerators are both

$$2u^3 + 14u^4 + 6u^5 + u^6.$$

At radius three, however, the minimal repairs are

$$\{145, 236\} \quad \text{for } G_{\text{dis}}, \qquad \{123, 256\} \quad \text{for } G_{\text{ov}}.$$

The first pair is disjoint and the second pair meets in helper 2. Inclusion–exclusion under homogeneous survival consequently gives

$$R^{(\leq 3)}_{\text{dis}}(s) = 2s^3 - s^6, \qquad R^{(\leq 3)}_{\text{ov}}(s) = 2s^3 - s^5.$$

The pointed polynomials agree while the filtered curves do not. □

Theorem 6.5 (Availability- and blocker-matched asymptotic separation)**.** *Over a finite prime field $\mathbb{F}_q$ there are two represented $[10, 4, 6]_q$ inner codes whose distinguished radius-three recovery data have the same pointed rank-triple multiplicity enumerator, five minimum repairs, locality three, matching number two, transversal number two, exactly one minimum blocker, and helper degree multiset*

$$(1, 1, 1, 1, 2, 2, 2, 2, 3).$$

Their homogeneous reliability polynomials are nevertheless

$$R^{(\leq 3)}_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \qquad R^{(\leq 3)}_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8.$$

Using one common outer family gives two asymptotically good fixed-alphabet families with the same length and dimension formulas, the same proved distance lower bound, and designated target classes of density exactly $1/10$ carrying these distinct bounded recovery data.

Proof. On helpers $H = \{0, \ldots, 8\}$ consider the two 3-uniform linear clutters

$$\mathcal{A} = \{158, 026, 045, 013, 478\},$$
$$\mathcal{B} = \{168, 237, 078, 015, 124\}.$$

Here 158 denotes $\{1, 5, 8\}$. Their helper degrees have the displayed common multiset. The pairs $\{0, 8\}$ and $\{1, 7\}$ are respectively their unique minimum transversals. Each clutter contains two disjoint edges, while a three-edge matching is impossible because every edge meets its two-point transversal. Hence both have matching and transversal numbers two and one minimum blocker.

Their different higher intersections are visible without a search. For each k, the table records the number of k-edge subfamilies by union size:

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9

Alternating these rows is inclusion–exclusion for the union of the five edge-containment events and gives the two stated reliability polynomials.

It remains to represent both clutters, and this is a generic geometric construction. Work first over an infinite field. For $\mathcal{A}$, choose five lines $\ell_1, \dots, \ell_5$ in a projective plane with ℓ_2, ℓ_3, ℓ_4 concurrent and otherwise in general position. Label their displayed intersections by

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5,$$

and choose $2, 6$ generically on ℓ_2, 3 on ℓ_4, and 7 on ℓ_5. For $\mathcal{B}$, make ℓ_1, ℓ_4, ℓ_5 concurrent at 1, put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose $6, 3, 5, 4$ generically on $\ell_1, \ell_2, \ell_4, \ell_5$. The choices can avoid the finitely many unwanted joining lines, so the listed edges are exactly the collinear triples in each configuration.

Choose representatives $p_i \in K^3$ of either plane configuration and put $x = (1, 0, 0, 0)$. Lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the three lifts above any listed line is one proper linear condition on the t_i, and dependence of any four helper lifts is also one proper linear condition, because every four quotient points span K^3. Avoiding this finite union of hyperplanes makes every helper triple and quadruple independent. The dependent four-sets containing x are then exactly $\{x\} \cup A$ with $A \in \mathcal{A}$ or $A \in \mathcal{B}$. Performing the construction over the rationals and reducing modulo one prime avoiding the finitely many nonzero determinants gives both representations over one finite prime field $\mathbb{F}_q$. Appendix A.2 records an explicit $\mathbb{F}_{29}$ instance as an independent cross-check, not as a premise of this existence proof.

Each resulting column matroid is rank-four sparse paving on ten elements, with five circuit-hyperplanes through x and none avoiding it. Thus the two pointed rank-triple multiplicity enumerators agree by Lemma 6.3. A hyperplane contains at most four columns and a displayed circuit-hyperplane contains four, so both row-span codes have parameters $[10, 4, 6]_q$. Their dual distance is four, and a displayed target circuit shows $\mu_x(0) = 4$; hence $z_x = 8$ for both seeds.

Put $L = \mathbb{F}_{q^4}$ and choose one asymptotically good L-linear outer family $O_N \leq L^N$ with dual distance tending to infinity. If its dimension and distance are K_N, D_N, the two concatenations have

$$n_N = 10N, \qquad k_N = 4K_N, \qquad d_N \geq 6D_N.$$

Since $3 + 1 < 8$, Theorem 4.1 eventually copies both the complete radius-three exact supports and normalized equations at every designated block target. Those N targets have density 1/10, and literal equality of both layers preserves all the matched clutter invariants and the two distinct reliability laws. □

Corollary 6.6 (Synthesis under exact recovery-structure transfer)**.** *Whenever Theorem 4.1 transfers a radius-r exact-support family and its normalized equations, it simultaneously transfers every smaller support filtration and every normalized decoder in that range. Consequently it preserves matching and transversal numbers, minimal blockers and their leading failure terms, multivariate reliability, and every bounded-EXIT curve through radius r. Any difference between two inner codes in one of these quantities therefore persists on their designated positive-density target classes.*

Proof. For each $t \leq r$, the radius-t data are obtained from the radius-r data by retaining precisely the witnesses with at most t helpers. The zero-extension bijection in Theorem 4.1 preserves supports, coefficients, and cardinality, so it commutes with every such truncation. Matching numbers, transversal numbers, and minimal blockers depend only on the resulting clutter. Reliability is the finite product-measure sum over its upward closure, the leading failure term is determined by its minimum blockers, and bounded EXIT is the complementary reliability under $s_v = 1 - p_v$. Each claimed consequence therefore follows from literal equality after relabeling. □

The retained field-nine harmonic computation gives a geometric instance of the same boundary: six-element circuits in a rank-five code add five-helper repairs beyond the classified radius-four supports. It illustrates the effect but is not evidence for Proposition 6.4.

7 Geometric flagships

7.1 The twisted-cubic–axis family

Let $q = 3^h$ and work in $\mathrm{PG}(3, q)$. Put

$$T_q = \{C(t) = (1 : t : t^2 : t^3) : t \in \mathbb{F}_q\},$$

$$L_q = \{A(u) = (0 : 1 : u : 0) : u \in \mathbb{F}_q\} \cup \{A_\infty = (0 : 0 : 1 : 0)\}.$$

The line L_q is the characteristic-three axis of the osculating-plane pencil [5]; all statements below follow directly from the displayed coordinates.

Theorem 7.1 (Cubic–axis recovery structure)**.** *The row code on $T_q \sqcup L_q$ is $[2q + 1, 4, q - 1]_q$. Its minimal circuits of size at most four are the triples on L_q and the sets*

$$\{C(s), C(t), C(u), (0 : e_1 : e_2 : 0)\}$$

for distinct s, t, u, where $e_1 = s + t + u$ and $e_2 = st + su + tu$. Every cubic coordinate has

$$(\nu, \tau) = \left(\frac{q-1}{2}, q - 2\right).$$

If $Z_3(q)$ is the maximum size of a subset of $(\mathbb{F}_q, +)$ containing no three distinct elements with sum zero, then

$$(\nu, \tau)_{A_\infty} = \left(\frac{5q-3}{6}, 2q - 1 - Z_3(q)\right),$$

$$(\nu, \tau)_{A(a)} = \left(\frac{5q-9}{6}, 2q - 2 - Z_3(q)\right).$$

In particular every coordinate satisfies $\tau > \nu$ for $q \geq 9$.

After adjoining $C(\infty) = (0 : 0 : 0 : 1)$, the completed row code is $[2q + 2, 4, q]_q$. Radius four exhausts its full minimal recovery-set family, and its rows are uniform on the two projective blocks:

$$\left(\frac{q-1}{2}, q - 1\right) \text{ on the cubic,} \qquad \left(\frac{5q-3}{6}, 2q - 3\right) \text{ on the axis.}$$

Proof. A plane containing L_q contains at most one cubic point by Frobenius injectivity; a plane not containing it meets the line once and the cubic at most three times. A plane through L_q and one cubic point has $q + 2$ columns, so the largest section has size $q + 2$; the length $2q + 1$ and rank four therefore give distance $q - 1$.

Write $\Delta(s, t, u) = (s - t)(s - u)(t - u)$. Direct expansion gives, up to the fixed ordering sign,

$$\det\bigl(C(s), C(t), C(u), (0 : a : b : 0)\bigr) = \Delta(s, t, u)(ae_2 - be_1).$$

Thus three distinct cubic points meet the axis in the unique point $(0 : e_1 : e_2 : 0)$. If $e_1 = 0$, then in characteristic three $e_2 = -(s - t)^2 \neq 0$, so this point is A_∞. Four cubic columns have nonzero Vandermonde determinant. Two distinct cubic and two distinct axis columns also have nonzero determinant, while any set containing three axis points already contains an axis triple. These calculations prove both the displayed circuit list and its completeness through size four.

We record the elementary hypergraph calculation used for the exact rows. Let $\mathcal{Z}$ be the triples of $\mathbb{F}_q$ with sum zero. The parallel class $\{a, a+1, a-1\}$ partitions $\mathbb{F}_q$, so $\nu(\mathcal{Z}) = q/3$; after deleting all triples through zero, deleting the one affected parallel-class member gives $\nu = q/3 - 1$. A set avoids $\mathcal{Z}$ exactly when it is three-term-zero-sum-free, hence $\tau(\mathcal{Z}) = q - Z_3(q)$. Translation preserves $\mathcal{Z}$ and lets a maximum avoiding set omit zero; adjoining the now-isolated zero shows that the deleted hypergraph has transversal number $q - 1 - Z_3(q)$.

At an axis target, the minimal clutter is the disjoint union of the complete graph on the other q axis points and a three-cubic component. At A_∞ that component is $\mathcal{Z}$. At $A(a)$, the bijection

$$s \longmapsto (s+a)^{-1}$$

identifies it with the zero-sum triples avoiding zero: expanding the numerator of the sum of the three inverses gives exactly $e_2 - ae_1 = 0$. Matching and transversal numbers add across the two disjoint helper blocks. Since K_q contributes $((q-1)/2, q-1)$, the two stated axis rows follow.

Fix now a cubic target $C(x)$. Relabel every other cubic parameter by $u = (s-x)^{-1} \in \mathbb{F}_q^*$. The repair using parameters s, t has axis completion

$$\chi_x(u+v), \qquad \chi_x(w) = \begin{cases} A_\infty, & w = 0, \\ A(-x + w^{-1}), & w \neq 0, \end{cases}$$

and χ_x is injective. If g generates $\mathbb{F}_q^*$, pairing g^{2i} with g^{2i+1} uses every nonzero parameter once and gives distinct colors $g^{2i}(1+g)$. This is a matching of size $(q-1)/2$, and no matching can be larger because every repair uses two cubic helpers. For the transversal, all but one cubic helper give a cover of size $q-2$. Conversely, if a cover selects t cubic helpers and leaves a nonempty set U unselected, fix $u_0 \in U$. For every $v \in U \setminus \{u_0\}$, the distinct axis color $\chi_x(u_0 + v)$ must be selected. Hence the cover has at least $t + (|U| - 1) = q - 2$ vertices. This proves the cubic row without an unstated complement lemma. The $q/3$ disjoint parallel-class triples force $Z_3(q) \leq 2q/3 \leq q - 2$ for $q \geq 9$, which gives $\tau > \nu$.

For the completion, projective shifted inversion $s \mapsto (s+a)^{-1}$, with its pole sent to infinity and infinity to zero, is induced by an invertible ambient map; it is transitive on each of the cubic and axis blocks. It therefore suffices to use targets $C(\infty)$ and A_∞. At $C(\infty)$ the radius-three repairs are $\{C(s), C(t), A(s+t)\}$. The same pairing and covering argument gives $((q-1)/2, q-1)$. At A_∞, the radius-three matching already has $(q-1)/2 + q/3 = (5q-3)/6$ edges. In any matching of minimal radius-four repairs, if an edge uses c cubic and a axis helpers, independence of the helper columns and the circuit classification give $2c + 3a \geq 6$: an axis-only edge has $a \geq 2$, a cubic-only edge has $c \geq 3$, and a mixed edge cannot have $c = 1$ or $(c, a) = (2, 1)$, since the corresponding sets are independent by the size-at-most-four classification. Every remaining mixed case satisfies the inequality. Summing over the disjoint edges and using the $q+1$ cubic and q available axis coordinates yields $6\nu \leq 2(q+1) + 3q = 5q + 2$, hence the same upper bound $(5q-3)/6$.

Finally, a helper set contains no full repair exactly when it lies in a hyperplane section avoiding the target. For $C(\infty)$, the largest such section has $q+2$ points (the plane $X_3 = 0$ attains it), giving $\tau = (2q+1) - (q+2) = q - 1$. A plane avoiding A_∞ meets the axis once and the cubic at most three times. To see sharpness, choose $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_3$. The plane through $C(0), C(1), C(\alpha)$ and their finite axis completion $(0 : e_1 : e_2 : 0)$, where $e_1 = 1 + \alpha \neq 0$, contains four displayed points and avoids A_∞. Hence $\tau = (2q+1) - 4 = 2q - 3$. Rank four makes every minimal circuit have at most five points, so radius four exhausts the full minimal recovery-set family. $\square$

This is the inner code in the strict weighted-transfer example above. The exact $q = 9$ rows are recorded in Appendix A.3.

7.2 The quartic–nucleus harmonic family

Let

$$V(t) = (1, t, t^2, t^3, t^4), \qquad V(\infty) = e_4, \qquad N = e_2,$$

and let $S_q = \{V(t) : t \in \mathbb{P}^1(\mathbb{F}_q)\} \cup \{N\}$. The classical normal-rational-curve nucleus formula identifies N as the hyperplane nucleus in characteristic three [6, Theorem 1].

Theorem 7.2 (Quartic–nucleus recovery structure)**.** *For $q = 3^h \geq 9$, the row code Q_q on S_q has parameters*

$$[q + 2, 5, q - 3]_q, \qquad d(Q_q^{\perp}) = 5.$$

Its minimal circuits of size at most five are exactly

$$\{N\} \cup B,$$

where B is a harmonic quadruple of $\mathbb{P}^1(\mathbb{F}_q)$. These quadruples form a Steiner system $S(3, 4, q+1)$, the characteristic-three four-set orbit in the standard $\mathrm{PGL}(2, q)$ design construction [21, Section 2]. Hence the radius-four repairs are all B at the nucleus and all $\{N\} \cup (B \setminus \{x\})$ at a curve target $x \in B$. The nucleus has

$$\frac{(q + 1)q(q - 1)}{24}$$

radius-four repairs, and every curve target has $q(q - 1)/6$. Every curve target has $(\nu, \tau) = (1, 1)$.

Proof. For distinct finite a, b, c, d, the hyperplane through their quartic columns contains N exactly when $e_2(a, b, c, d) = 0$; with ∞ present the condition is $a + b + c = 0$. Given three finite points, if $e_1 = a + b + c \neq 0$, the unique completion is $d = -e_2/e_1$; if $e_1 = 0$, the unique completion is ∞. In the first case, substituting $d = a$, for example, changes the harmonic equation to $(a - b)(a - c) = 0$, and similarly for b and c; hence the completion is distinct. In the second case, writing $c = -a - b$ gives $e_2(a, b, c) = -(a - b)^2 \neq 0$, so no finite completion exists. A triple $\{\infty, a, b\}$ has the unique completion $-a - b$, distinct from a, b in characteristic three. This proves the Steiner property. Since $\{\infty, 0, 1, -1\}$ is one block and $\mathrm{PGL}(2, q)$ is 3-transitive, unique completion also identifies the blocks with its projective orbit.

Any five curve points are independent by the extended Vandermonde determinant. Expanding the determinant of $N, V(a), V(b), V(c), V(d)$ along the nucleus row gives

$$\Delta(a, b, c, d)e_2(a, b, c, d),$$

and with infinity present it gives $\Delta(a, b, c)e_1(a, b, c)$. The Vandermonde factor is nonzero for distinct parameters. Thus $\{N\} \cup B$ is dependent exactly for a harmonic block. A complementary minor shows that N with any three distinct curve points is independent; in the zero-sum finite branch its determinant reduces to $-\Delta(a, b, c)e_2(a, b, c)$, which is nonzero by the calculation above. Deleting any point therefore restores independence. These are all circuits of size at most five, so the dual distance is five.

A nonzero hyperplane restricts to a nonzero polynomial of degree at most four on the normal rational curve, counting infinity as a root when the quartic coefficient vanishes. It therefore contains at most four curve points. If it also contains N, equality occurs exactly on a harmonic block, so the largest section of S_q has five points. Hence the primal distance is $(q + 2) - 5 = q - 3$. Finally, the Steiner count is $\binom{q+1}{3}/\binom{4}{3}$, and the number of blocks through a fixed curve point is $\binom{q}{2}/\binom{3}{2}$. The circuit–repair correspondence gives the stated recovery sets and counts. $\square$

The inner-dual half of the coarse transfer gate is automatic at radius four because $5 < 2d(Q_q^\perp) = 10$. Theorem 4.1 therefore places the entire harmonic recovery structure with positive density in asymptotically good fixed-$\mathbb{F}_q$ families.

7.3 Reliability, EXIT, and the geometric contrast

The nucleus repairs are the blocks of the Steiner system $S(3, 4, q + 1)$. At a curve target, every radius-four repair contains the nucleus. Thus the nucleus recovery structure is a parallel family of four-helper repairs, whereas the curve recovery structure has a compulsory series helper. The reliability and bounded-EXIT theorem applies to both target types without an exact finite profile. Exact field-nine profiles are computational refinements recorded in Appendix A.3; they are not used in the body theorem chain.

Finally, the harmonic small-circuit closure has a deterministic nucleus gate. Let $D(S)$ close curve set S under "three points of a harmonic block add the fourth." Then

$$\mathrm{cl}_4(S) = S \quad \text{if } S \text{ contains no block}, \qquad \mathrm{cl}_4(S) = \{N\} \cup D(S) \quad \text{otherwise},$$

and $\mathrm{cl}_4(S \cup \{N\}) = \{N\} \cup D(S)$. Indeed, without N no curve point can be repaired, while a complete block $B \subseteq S$ repairs N. Once N is present, the curve-target repairs are exactly N together with three points of a harmonic block, so iteration is precisely harmonic completion. This proves all three identities.

Appendix A.3 gives a field-nine block-free rank-five spanning set and its one-round completion after adjoining N. These witnesses illustrate the nucleus gate but are not inputs to the closure identities, an asymptotic propagation claim, or a threshold theorem.

8 Conclusion

At a target coordinate, exact helper supports, their upward-closed recovery-set family, and normalized recovery equations give three views of bounded local recovery. Locality and overlap statistics summarize the recovery-set family, which is generated by the support projection of the equations; each passage can lose information. The minimal clutter measures availability and adversarial tolerance, the coefficients give exact scalar recovery and the weighted transfer cost, and the Boolean event gives reliability and bounded EXIT. Exact concatenation transfers the supports and equations. Its persistent pointed obstruction is necessary and sufficient for eventual confinement in the fixed-inner construction and permits positive-density realization below the gate.

The cubic–axis family shows rich overlapping recovery sets and strict weighted transfer beyond a support-only gate. The quartic–nucleus family replaces that overlap pattern by a harmonic Steiner design: nucleus repairs form a parallel block family while curve targets share a compulsory series helper. Pointed-Tutte theory explains the full-radius structure; the radius filtration records when each repair first becomes available.

The structural $[10, 4, 6]_q$ pair makes this boundary persist after the usual coarse explanations have been removed. Its pointed-perspective data, repair count, locality, matching and transversal numbers, unique minimum-blocker count, and helper degree multiset all agree; only the higher-order overlap pattern changes the reliability law. A common outer family matches the global parameter formulas while exact transfer repeats the two distinct laws on coordinate classes of density $1/10$. The field-seven pair remains the smaller explicit precursor. This is the paper's central synthesis: represented seed structure, the weighted transfer gate, positive-density realization, and stochastic behavior form one theorem chain.

A sharper cubic blocker-stability statement is not needed for the present results. Sequential composition, general service regions, coefficient optimization, log-concavity, and random harmonic-cascade thresholds lie outside this paper.

A Formal verification and finite evidence

The recovery-equation reconstruction and exact transfer chains are checked in Lean. The import-only gate `RepairPorts.Gates.CompletePorts` audits the paper-facing declarations. Its two transfer terminals are

`RepairPorts.exactFunctionalStrata`,
`RepairPorts.exactPointedConfinementAndTransfer`.

The first gives the exact zero-, singleton-, and multisupport-functional profiles; the second gives the pointed minimum formula and exact equality of the bounded helper supports and normalized equations. Both report only `propext`, `Classical.choice`, and `Quot.sound`. The strict field-nine corollary is the conclusion of

`RepairCodes.projectiveAxisTwistedCubic_strict_weighted_transfer_of_regular_projective_action`.

Singer regularity is an explicit hypothesis of that declaration, not a hidden formal axiom.

The completed cubic parameters and rows are independently exposed by

`FiniteGeom.projectiveAxisTwistedCubic_code_parameters`,
`RepairCodes.minimalProjectiveAxisTwistedCubicRepair_full_eq_four`,
`RepairCodes.minimalProjectiveCubicRepair_four_invariants`,
`RepairCodes.minimalProjectiveAxisRepair_four_invariants`.

The gate also prints the corresponding full-radius invariant terminals.

The paper's `verification/formal-boundary.json` fixes the formal source commit, the finitegeom base and release commits, Lean and mathlib revisions, 36-module closure, and 61 paper-facing terminals:

source `bb3d7c34b0381ee1ba77ad95076959e53f742d19`,
finitegeom base `e69fbe7ac9e2dd6fb193f6f3c914d3831cbe806e`,
finitegeom release `36c83268ddaeec9ee22824cad44d6222a9e67081`,
gate fact SHA-256 `fbb29ef0bc559b9ac2ce3308a7b3c0572753ffd7163d6c15cd18e57464fc0a4d`.

The immutable area-export planner reports 888036 closure bytes and no prose drift at those commits. The companion `verification/README.md` gives the guarded build, axiom-audit, and area-export replay commands. These identifiers distinguish the kernel-checked module closure from the manuscript PDF and from evidence-only finite calculations. The named `verification/` files are supplementary files distributed with the paper's source archive.

Trace duality, scaled concatenated parameters, exact designated-class density, eventual necessity and sufficiency, support/coefficient transfer, and the MDS fingerprint corollary are audited through separate declarations. In particular, the paired declaration below covers only simultaneous eventual radius-three support/coefficient transfer through one common outer family; it does not encode the quotient-plane seed construction, the resulting $[10, 4, 6]_q$ representations, their finite clutter invariants, or the human assembly of Theorem 6.5:

`RepairPorts.concatenatedRestrictedCode_parameters`,
`RepairPorts.eventually_pointedConfinement_iff_zeroCost`,
`RepairPorts.eventually_prescribedPorts`,
`RepairPorts.eventually_radiusThree_prescribedPortPair`,
`RepairPorts.eventually_mdsMinimumCoefficientFingerprints`.

They report only `propext`, `Classical.choice`, and `Quot.sound`. Random-GV or AG/TVZ existence of an outer family is the explicit classical input; no such existence theorem occurs in the Lean dependency closure.

The finite reliability and bounded-EXIT calculus is audited through

`RepairPorts.portReliability_delete_contract`,
`RepairPorts.hasDerivAt_portReliability_update`,
`RepairPorts.hasDerivAt_homogeneous_portReliability`,
`RepairPorts.erasureFailureProbability_delete_contract`,
`RepairPorts.noRepairProbability_eq_erasureFailure`,
`RepairPorts.cheapestRepairRadiusProbability_eq_failure_sub`,
`RepairPorts.blockerFailurePolynomial_eq_minimum_term_add_remainder`.

These declarations use finite sums and polynomial algebra only. Exact finite profiles and Poisson approximations remain computational refinements outside the body proof chain.

For the concrete completed field-nine family, the external existence input is Stichtenoth's self-dual TVZ-family theorem over $\mathbb{F}_{6561}$ [19, Theorem 1.6(ii)]. Classical inputs are labeled as such: outer-family existence, concatenated-dual decomposition, Singer regularity, normal-rational-curve nuclei, and the Las Vergnas perspective polynomial.

A.1 The two field-seven seed representations

For completeness, order the four-subsets of $\{0, \ldots, 6\}$ lexicographically as

$$\begin{gathered}0123, 0124, 0125, 0126, 0134, 0135, 0136, 0145, 0146, 0156,\\ 0234, 0235, 0236, 0245, 0246, 0256, 0345, 0346, 0356, 0456,\\ 1234, 1235, 1236, 1245, 1246, 1256, 1345, 1346, 1356, 1456,\\ 2345, 2346, 2356, 2456, 3456.\end{gathered}$$

The corresponding determinants in $\mathbb{F}_7$ are

$$\begin{aligned}G_{\mathrm{dis}}: {}& 5, 3, 3, 5, 4, 4, 1, 0, 5, 5, 2, 5, 0, 6, 5, 2, 1, 1, 6, 4,\\ & 1, 6, 6, 3, 4, 6, 4, 6, 5, 2, 0, 1, 6, 3, 4,\\ G_{\mathrm{ov}}: {}& 0, 2, 5, 1, 2, 5, 1, 5, 2, 6, 1, 6, 4, 4, 5, 0, 5, 4, 4, 5,\\ & 1, 6, 4, 2, 2, 4, 3, 1, 1, 3, 6, 2, 2, 6, 0.\end{aligned}$$

Each triple extends to a displayed nonzero minor, and each set of at least five columns contains one. This proves the finite representation facts used in Proposition 6.4. The independent exact replay bundle is

`verification/f7-seed.py`,
`verification/f7-seed.json`.

It checks the circuit-hyperplane lists, pointed rank-triple multiplicity enumerator, minimal radius-three repairs, and survivor-set counts. Its certificate SHA-256 is

`8096230e66f634c820ae7ec4bacd9b2493006782ff02b8be3a8c7e1caf80de07`.

A.2 Concrete matched-seed cross-check

The proof of Theorem 6.5 is the generic line-and-lift argument in the body and does not depend on a finite search. For an exact replayable instance, the supplementary bundle

`verification/matched-seed.py`,
`verification/matched-seed.json`

records two 4×10 matrices over $\mathbb{F}_{29}$. It independently checks rank four by elimination, every four-column dependence both by elimination and by the determinant formula, the complete circuit-hyperplane lists, parameters $[10, 4, 6]_{29}$ and dual distance four, equality of the complete pointed rank-triple histograms, the matching, transversal, blocker, and degree data, and the two reliability polynomials. Its certificate SHA-256 is

`16a378edc882a6dda7f5642c7d21bfa49913b45ef5409398de117897b19dd2b8`.

This concrete instance cross-checks the structural proof; it is not a premise of the theorem.

A.3 Field-nine evidence

For the uncompleted field-nine cubic–axis code on $T_9 \sqcup L_9$, $Z_3(9) = 4$, and the cubic, finite-axis, and axis-infinity coordinate types have exact radius-three (ν, τ) rows

$$(4, 7), \qquad (6, 12), \qquad (7, 13),$$

with multiplicities $9, 9, 1$. These are not rows of the completed code. After adjoining $C(\infty)$, Theorem 7.1 gives the two uniform radius-four/full-radius recovery-set rows

$$(4, 8) \quad \text{on the ten cubic coordinates}, \qquad (7, 15) \quad \text{on the ten axis coordinates}.$$

Corollary 3.2 uses the completed 20-coordinate code. Its generalized-SPC check has coefficients $(1, a, 1, 1, 1)$, where a is a linear Singer multiplier whose translate of the 20 unit-cost projective functional classes is disjoint from the original 20-set. Regularity on 820 classes and $20^2 < 820$ prove that such an a exists. Thus this is an exact conditional-on-Singer existence proof, not a claim that five field elements or a hidden finite multiplier certificate were computed. The functional support distance is five, both the weighted nonzero sector and the exact pointed nonembedded threshold are at least six, and radius-four transfer follows.

For the quartic–nucleus code at $q = 9$, the general formulas give 30 nucleus repairs and 12 repairs at each curve target. The certificate additionally gives nucleus values $(\nu, \tau) = (2, 5)$, exact Bernstein and EXIT rows, and finite propagation witnesses. In particular, a block-free rank-five spanning set completes in one round after adjoining the nucleus. These exact outputs are evidence-only refinements: no theorem in the main body depends on them.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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